Decoding Two Strike Pitch Sequencing in MLB

Advanced Sports Performance Analytics: Industry Report

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EXECUTIVE SUMMARY

Two-strike counts represent the most critical moments of a plate appearance in Major League Baseball. Often, the decision of which pitch to throw is influenced by the previous pitch, with the goal of deceiving the batter through changes in velocity, movement, and location. This is regularly referred to as 'pitch tunneling' because the two different pitches first appear to be arriving on the same trajectory, or tunnel, before diverging. The purpose of this report is to analyze pitch sequencing strategies in two-strike counts to identify the patterns and metrics that most effectively result in strikeouts. Analyzing over 28,000 two-strike pitches from the 2026 season, our key finding is that the current pitch characteristics — specifically location, fastball shape, and vertical/horizontal movement — are the primary drivers of strikeouts, heavily overriding the differential changes from the preceding pitch. The dataset exhibited a baseline strikeout rate of 19.4%. Considering these findings, we recommend that pitchers focus on pitch location and trusting their arsenal rather than over-relying on specific sequencing strategies. Front offices should prioritize drafting and developing pitchers with elite velocity and fastball shapes, cultivating command in the minor leagues when given the opportunity to work alongside professional coaches and front offices.

INTRODUCTION / BUSINESS PROBLEM

In the modern era of MLB, front offices and coaching staffs rely heavily on advanced analytics to gain a competitive edge. With the advent of Statcast tracking technology, teams have access to highly granular data on every pitch thrown. Maximizing the probability of a strikeout in a two-strike count directly impacts run prevention and overall game outcomes, making it a crucial research area. The primary stakeholders for these insights include General Managers optimizing draft and development pipelines, coaching staffs devising in-game strategies, and pitchers executing on the mound. The objective of this analysis is to identify distinct pitch sequencing patterns in two-strike counts and evaluate how changes between consecutive pitches influence the likelihood of a successful outcome (a strikeout). Going into this study, our primary hypothesis was that pitchers who create greater

differences between consecutive pitches in velocity, movement, and location would generate more successful outcomes by amplifying the tunneling effect.

DATA

The data for this analysis was sourced from MLB's Statcast website, collected publicly via the Python `pybaseball` package. The dataset encompasses pitch-level tracking data for approximately the first month of the 2026 season (March 25 – April 20). The original dataset contained over 100,000 pitches and 118 features. For this study, the data was filtered exclusively to pitches thrown in two-strike counts, yielding a sample size of over 28,000 rows. The primary Key Performance Indicator (KPI) and dependent variable is a binary indicator of whether the pitch resulted in a strikeout. The dataset features 38 independent variables, including absolute metrics like pitch velocity, horizontal and vertical movement, location relative to home plate, arm slot, and spin rate. To test sequencing, we engineered additional features calculating the differentials from the previous pitch in the same at-bat. Furthermore, all pitches were standardized to a right-handed perspective to normalize the analysis across left-handed and right-handed matchups. A key limitation of the data is the restricted early-season timeframe, which may be influenced by cold weather or early-season behavioral/strategical variance.

METHODOLOGY

To parse this multidimensional dataset, we deployed a suite of machine learning and statistical methods, prominently featuring a LightGBM classification model. LightGBM was selected for its robustness in handling large, tabular datasets and its ability to natively compute Feature Importances and SHAP (SHapley Additive exPlanations) values, offering transparency into what variables drive the model's predictions. We tested the predictive power of absolute pitch metrics versus pitch-to-pitch differentials to evaluate the tunneling hypothesis. A paired t-test was used to validate the statistical significance of these variables. Additionally, we utilized Principal Component Analysis (PCA) to reduce the dimensionality of the pitch characteristics. This was followed by Factor Analysis (FA), which identified underlying latent constructs driving pitcher behavior. Finally, K-Means Clustering was applied to the principal components to uncover natural groupings in pitch execution strategies without relying on predefined labels. This multi-layered approach ensures that the model structure is both mathematically rigorous and practically interpretable for baseball personnel.

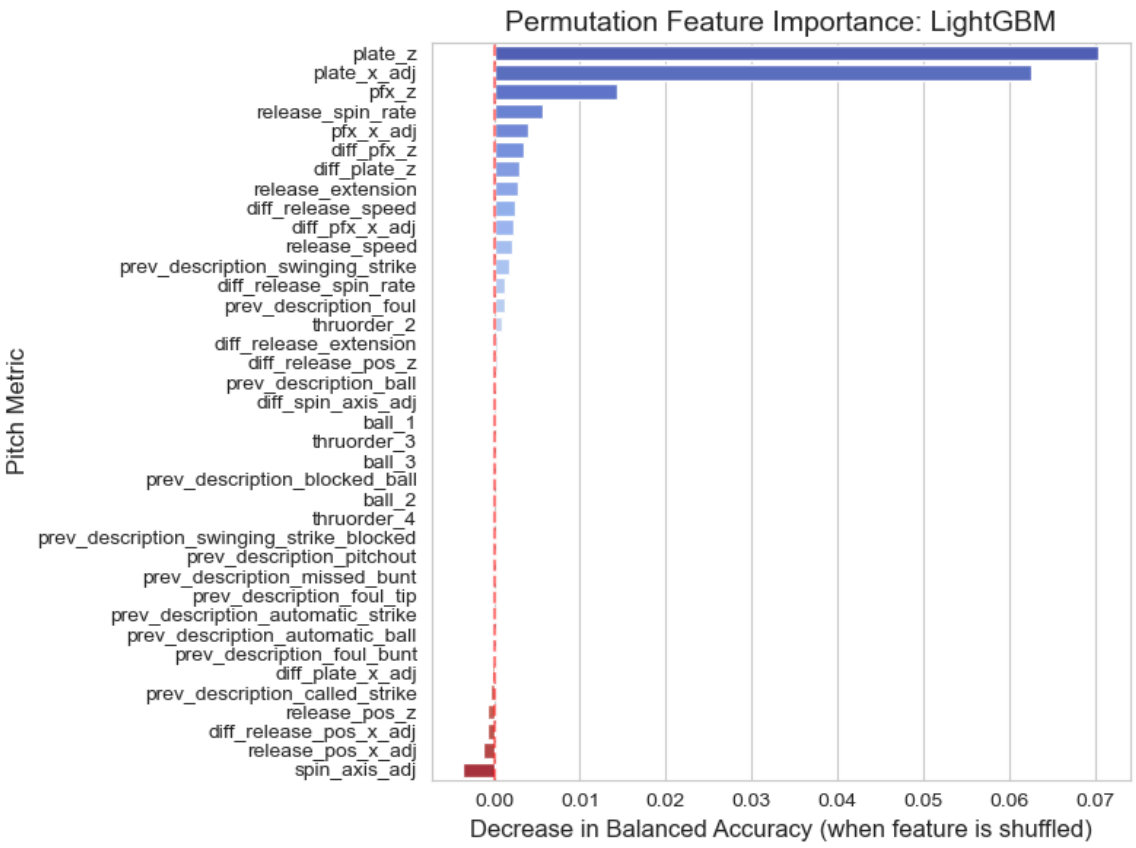
RESULTS

The results of the LightGBM classifier strongly challenged our initial hypothesis. Feature importances and SHAP values revealed that absolute metrics—specifically pitch location across the plate, fastball shape, and vertical/horizontal movement—were the dominant predictors of a strikeout. Conversely, differential metrics (the change in velocity, location, or

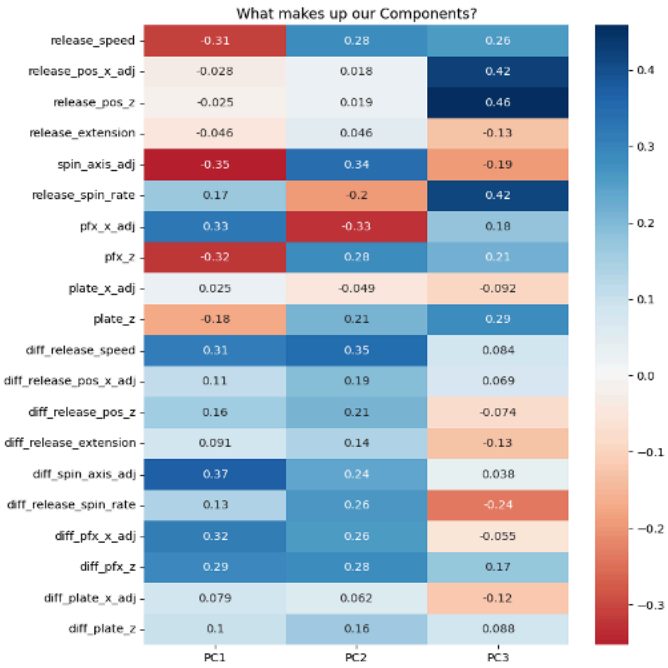
movement from the preceding pitch) ranked lower in predictive power. The paired t-test confirmed these results, as the balanced accuracy score of the full LightGBM model was not statistically different from the model without differential metrics. Our Factor Analysis distilled the complex feature set into four primary latent factors: (1) Pitch Shape & Movement Profile, (2) Pitch-to-Pitch Adjustments, (3) Vertical Approach / Power Profile, and (4) Release Mechanics / Arm Slot. Applying K-Means Clustering to the PCA-reduced data (visualized in 2D) demonstrated clear separations along the primary components of shape and location. Ultimately, statistical significance testing confirmed that executing a high-quality pitch to an optimal location fundamentally outweighs the deceptive benefits of sequencing differentials.

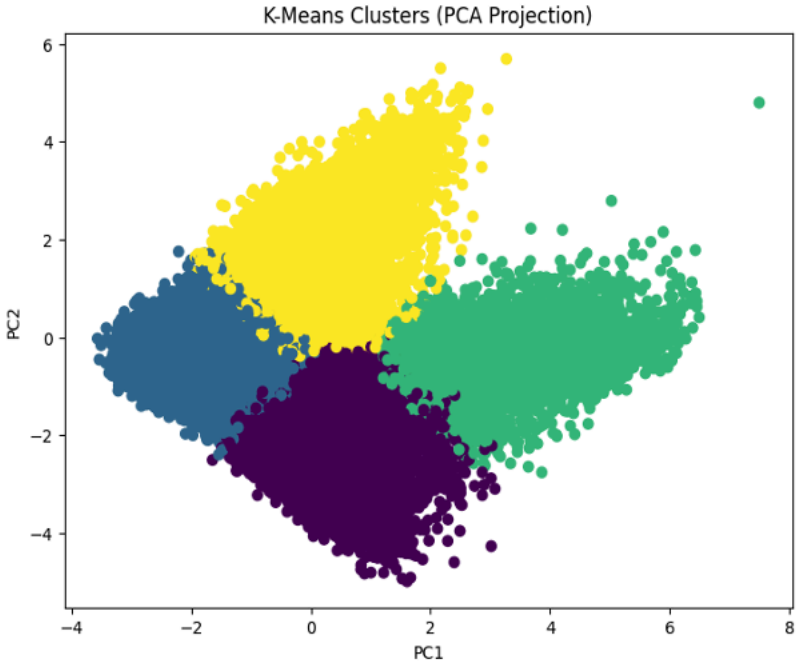
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=== CLASSIFIER RESULTS (sorted by Balanced Test Accuracy) ===
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	CV Mean	CV Std	Balanced Test Accuracy
LightGBM	0.6132	0.0022	0.6289
Hist Gradient Boosting	0.6127	0.0059	0.6276
Decision Tree (Pruned)	0.6049	0.0089	0.6144
Random Forest	0.5982	0.0085	0.6095
XGBoost	0.5803	0.0029	0.5971
Bagging	0.5852	0.0058	0.5819




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Full model 10-fold CV Balanced Accuracy score: 0.6088
Model without previous pitch 10-fold CV Balanced Accuracy score: 0.6144
Mean Balanced Accuracy Improvement: -0.0055
P-value: 0.1561
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DISCUSSION

While pitch tunneling and sequencing remain popular concepts, the performance data implies that front offices and coaches may be overvaluing the relationship between consecutive pitches relative to the raw quality of the pitches themselves. It was highly unexpected that pitch-to-pitch differentials did not rank more highly in the LightGBM model.

One might assume that an 85 mph changeup following a 98 mph fastball generates whiffs primarily because of the 13 mph differential. However, the model indicates that the intrinsic shape and precise location of that changeup do the majority of the work. In the context of baseball strategy, these results validate the traditional belief that elite stuff and command supersede trickery. A perfectly executed pitch is difficult to hit, regardless of what the batter saw on the previous pitch.

PRACTICAL IMPLICATIONS

These findings yield actionable recommendations across the baseball operations. For General Managers and their scouting departments, the mandate is clear: prioritize drafting and acquiring players with elite raw fastball velocity and premium pitch shapes. Command should be the primary developmental focus as players enter the minor leagues and engage with professional coaching staffs. For the coaching staff, game planning should focus heavily on identifying optimal locations for a pitcher's arsenal based on their pitch shapes, arm angles, and opponent's weaknesses. While calling a good sequence is helpful, coaches must drill into their pitchers that hitting the target and trusting their stuff is the ultimate priority. For pitchers, the offseason is the time to build velocity and refine pitch shapes. In-game, the psychological focus must narrow entirely to executing location and trusting the catcher's calls. Furthermore, pitchers should recognize that generating weak contact on a two-strike pitch is still a highly successful outcome, mitigating the pressure to force a strikeout through perfect sequencing.

FUTURE DIRECTIONS

To scale these findings, MLB organizations should implement these insights directly into advance scouting, shifting the weight of pre-game reports slightly away from sequencing suggestions and more toward absolute location vulnerabilities. Future research must expand the dataset beyond a single month to encompass multiple full seasons. This will help determine if these findings hold stable across varying offensive environments, weather conditions, late-season pitcher fatigue, and playoff environments. Additionally, extending the model to incorporate biomechanical hitter profiles and bat tracking data (e.g., bat speed and swing path) would provide a more holistic view of why certain pitch shapes succeed against specific swing archetypes.

LIMITATIONS

This study is subject to several data and methodological limitations. First, pitch-level metrics are generally better suited for predicting 'whiffs' (a.k.a. swinging strikes) rather than strikeouts, as a strikeout is a situational outcome dependent on the count. Second, the hitter matchup is a massive, complex confounder that the model cannot not fully encapsulate. A pitcher can execute a pitch with optimal shape and location, but an elite hitter may still spoil it. Furthermore, our sample is constrained to early-season data, which

may introduce biases related to cold weather suppressing offensive output or pitchers building early-season arm strength. Finally, from a validity standpoint, a pitcher's ultimate goal is run prevention, not strictly accumulating strikeouts. By isolating the strikeout as the sole success metric, we exclude other highly positive outcomes like weak ground balls or popups.

CONCLUSION

In summary, this analysis of two-strike pitch sequencing reveals that absolute pitch quality—defined by location, velocity, and movement—is substantially more predictive of strikeouts than the differentials between consecutive pitches. This finding holds immense decision relevance for MLB front offices, suggesting that developmental resources and draft capital should be aggressively allocated toward raw 'stuff' and command rather than sequencing aptitude or an ideal pitch mix. Moving forward, teams should simplify the psychological burden on pitchers in two-strike counts: trust your best pitch, hit your spot, and let the physics of the pitch generate the out.

DATA DICTIONARY

Statcast Variables

- **pitch_type**: The type of pitch thrown.
- **game_date**: The date of the game.
- **player_name**: The name of the pitcher.
- **at_bat_number**: The sequential number of the at-bat within the game.
- **pitch_number**: The sequential number of the pitch within the at-bat.
- **strikes**: The number of strikes in the count before the pitch.
- **events**: The outcome of the at-bat (e.g., strikeout, single, field_out).
- **description**: The result of the specific pitch (e.g., called_strike, foul, hit_into_play).

Adjusted Performance Metrics

These variables were modified to standardize the data, making Left-Handed Pitchers (LHP) and Left-Handed Hitters (LHH) comparable to Righties.

- **release_pos_x_adj**: Horizontal release point, adjusted so that all pitchers appear as right-handed.
- **release_pos_z**: Vertical release point.
- **release_speed**: The velocity of the pitch in MPH at the point of release.
- **release_extension**: The distance from the rubber toward home plate where the ball is released.
- **spin_axis_adj**: The spin axis of the pitch, mirrored for LHP to maintain a consistent directional scale.

- **release_spin_rate**: The rotation speed of the ball in RPM.
- **pfx_x_adj**: Horizontal movement (inches), adjusted for pitcher handedness.
- **pfx_z**: Vertical movement (inches).
- **plate_x_adj**: The horizontal location of the pitch at the front of the plate, adjusted so that positive values are always outside and negative values are inside for all batters.
- **plate_z**: The vertical height of the pitch at the front of the plate.

Engineered Sequence Features

These variables capture the relationship between the current pitch and the pitch immediately preceding it in the same at-bat.

- **prev_[metric]**: The value of the specific metric (e.g., speed, movement) from the pitch immediately prior.
- **diff_[metric]**: The calculated change between the current pitch and the previous pitch (e.g., diff_release_speed represents the velocity gap).
- **prev_description_[result]**: Indicators of what happened on the previous pitch (e.g., was the previous pitch a foul or a called strike).

Contextual & Target Variables

- **ball_1, ball_2, ball_3**: Binary indicators for the number of balls in the count.
- **thruorder_[n]**: Indicators for how many times the pitcher has faced the batting order in that game.
- **is_strikeout**: Target variable, a binary indicator (1 if the pitch resulted in a strikeout or strikeout-double-play, 0 otherwise).